

CSE 370 – Database Systems

Theory Section 07 / 08 | Assignment 02

Spring 2023

Marks

___ / 20

Full Name (in Block Letter): _____ ID: _____

Question 1 [CO1] : 20 Points

Suppose that each of the following update operations (a-j) is applied directly to the database state shown in Figure A2_1. Discuss all integrity constraints violated by each operation (check Figure A2_2 for Referential Integrity constraints on the database of Figure A2_1). If any, provide different ways of enforcing these constraints.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Figure A2_1 : COMPANY database.

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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Figure A2_2 : Referential integrity constraints on the COMPANY database of Figure A2_1.

Consider each of the following operations (a-j) independent of each other. Each operation will be applied directly to the database state shown in Figure A2_1. For example, when you will consider no. b, you should not consider whether no. a has successfully been completed.

a. Insert <'Gavin', 'F', 'Harrison', '943775543', '1972-06-21', '2365 Newcastle Rd, Bellaire, TX', 'M', 58000, '888665555', 1> into EMPLOYEE.

Answer:

b. Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT.

Answer:

c. Insert <'Production', 4, '943775543', '2007-10-01'> into DEPARTMENT.

Answer:

d. Insert <'677678989', NULL, '40.0'> into WORKS_ON.

Answer:

e. Insert <'453453453', 'John', 'M', '1990-12-12', 'Spouse'> into DEPENDENT.

Answer:

f. Delete the WORKS_ON tuples with Essn = '333445555'.

Answer:

g. Delete the EMPLOYEE tuple with Ssn = '987654321'.

Answer:

h. Delete the PROJECT tuple with Pname = 'ProductX'.

Answer:

i. Modify the Mgr_ssn and Mgr_start_date of the DEPARTMENT tuple with Dnumber = 5 to '123456789' and '2007-10-01', respectively.

Answer:

j. Modify the Super_ssn attribute of the EMPLOYEE tuple with Ssn = '999887777' to '943775543'.

Answer: